

Cross-Laboratory Validation of EEG Classification of Reported Dream Experience

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Abstract

Electroencephalographic classifiers of reported dream experience may exploit laboratory structure even when participants are held out. We audited the released DREAM Figure 5 workflow and analyzed 1,065 strict binary reports from five collections (730 NREM; 335 REM). A close adaptation approximately recovered record-wise discrimination (NREM spectral power, AUROC = 0.569; REM filtered Catch22, AUROC = 0.636). With participant-held-out predictions pooled across laboratories, AUROC was 0.534 [0.500, 0.569] and 0.629 [0.567, 0.701], respectively. Restricting concordant pairs to the same laboratory reduced both primary estimates to approximately chance (0.498 [0.462, 0.536] and 0.498 [0.440, 0.576]). A classifier using only training-fold laboratory prevalence attained pooled AUROCs of 0.564 and 0.687. Cross-laboratory comparisons constituted 69.4% of NREM and 70.0% of REM pooled positive-negative pairs. When a whole laboratory was held out, primary performance was 0.493 and 0.470. Conversely, laboratory identity was readily decoded from published feature families (AUROC = 0.808–0.953). Within-laboratory REM performance was estimator- and collection-dependent, including a fold-conditioned sensitivity of 0.552 and 0.563 after excluding the severely imbalanced Turku collection. Across the tested DREAM collections, the released workflow supports pooled discriminability but does not establish a common, transportable EEG marker of reported dream experience.

1 Introduction

Dream reports pose a distinctive measurement problem: the target experience is private, and its label is observed only after awakening. The DREAM database harmonized multiple EEG and mentation collections and reported that spectral power and Catch22 features discriminated “Experience” from “No experience” under pooled random splitting [1, 2]. This is an important benchmark, but a pooled multicenter AUROC is not a single-domain quantity.

Participant separation and laboratory separation answer different scientific questions. Participant-held-out validation tests whether a rule generalizes to new peo-

ple while acquisition domains remain represented on both sides of the split. Laboratory-held-out validation tests transport to a new acquisition and reporting domain. Between those levels lies a diagnostic estimand: site-conditioned AUROC, which asks whether held-out participants with and without reported experience are ranked correctly when both records came from the same laboratory. If label prevalence and score location differ by site, pooled AUROC may be high even when within-site ranking is absent. This is a form of domain confounding distinct from direct participant leakage [3, 4].

Prior dream-decoding studies include blinded spectral-power tests and single-collection classifiers [5, 6], but our focused search did not identify an earlier laboratory-held-out evaluation of the exact DREAM Figure 5 workflow. We therefore performed a reproducibility and falsification analysis with five levels: record-wise discrimination, participant-held-out pooled discrimination, participant-held-out site-conditioned discrimination, laboratory-held-out transport, and laboratory-identity and prevalence-only controls. We also retained two mechanistically motivated extensions—Riemannian covariance and temporal irreversibility—with alignment and spectral-surrogate controls [7–9].

2 Materials and Methods

2.1 Source audit and analytic sample

We retrieved the versioned public collections enumerated by the deposited Figure 5 code and verified archives against publisher-provided sizes and MD5 digests. DREAM code 2 (Experience) was mapped to 1 and code 0 (No experience) to 0. Experience without recall, ambiguous or unknown labels, and wake observations were excluded. Subject identifiers were scoped jointly by source collection and original identifier. Three Young Adults EDFs fail CRC despite the archive matching its published MD5, and one Turku EDF is absent from its archive; these records were excluded.

The paper describes six collections across its broader EEG analyses, whereas the Figure 5 classifier lists five. We did not add Oudiette N1Data post hoc: its sleeping subset contains only one strict No-experience report, and its Fp1/C3/O1 montage does not match the classi-

fier’s F4/C4/O2 montage. The final code-path sample contained 1,065 reports from 166 unique participants. Because prevalence is part of the confounding question, Table 1 reports counts separately by sleep stage and collection.

2.2 Reproduction boundary and features

Literal execution of the deposit is blocked by three documented gaps. It imports a missing `feat_ext_funcs` module; it specifies a semicolon delimiter for a public comma-delimited records file; and it expects Siclari `formatted_*.edf` files and F4-M1/C4-M1/O2-M1 channels absent from all public versions inspected. We computed filtered Catch22 directly with `pycatch22` and reconstructed Siclari F4/C4/O2-M1 from the nearest HydroCel-256 sensors. These adaptations permit conceptual, not exact numerical, reproduction.

We analyzed the final 30 s before each report and resampled signals to 128 Hz for time-series features. Published families comprised normalized multitaper power in six bands across three channels, broadband Catch22, and Catch22 after six band-pass filters. Covariance matrices were trace-normalized, shrunk by one percent, and represented in a log-Euclidean tangent space. A label-free transductive alignment subtracted each laboratory-stage log-Euclidean mean. Temporal irreversibility used ordinal-pattern Jensen–Shannon divergence and a cubic time-asymmetry statistic at four lags. Twenty-four deterministic phase-randomized surrogates per channel provided a spectrum-preserving control.

2.3 Models and validation estimands

Binary models used the released XGBoost settings: learning rate 0.1, depth 3, subsampling 0.8, feature subsampling 0.8, and the workflow’s implicit default of ten boosting rounds [10]. No feature-family-specific optimization was performed. Random-record and participant-held-out analyses used five folds and 20 deterministic shuffled repetitions. Participant folds were generated by stratified group cross-validation, with no participant in both train and test partitions.

For each repetition, *pooled participant-held-out AUROC* compared every out-of-fold positive with every out-of-fold negative, regardless of source. *Site-conditioned AUROC* retained only positive–negative pairs from the same laboratory. We report both a macro average giving each eligible laboratory equal weight and a pair-weighted value giving each within-site pair equal weight. *Participant-equal* estimates assigned each participant a total weight of one before forming pairs. *Laboratory-held-out AUROC* trained on all other collections and evaluated the untouched target; its summary is the unweighted mean across four eligible target laboratories per stage.

A site-only baseline assigned each test record the experience prevalence of its laboratory estimated from the

corresponding training fold; no test labels were used. Because training prevalence varies inversely with held-out fold composition, direct pooling across mutually exclusive test folds can induce an artificial within-site ordering even for this constant-within-fold baseline. After observing this evaluation artifact, we added a transparently labeled estimator clarification: fold-conditioned site AUROC, which restricts pairs to the same laboratory *and* test fold. The site-only baseline is exactly 0.5 under this estimand. This sensitivity was not used to select models.

AUROC was decomposed exactly into concordant within-site and cross-site pair contributions. Uncertainty used 2,000 participant-cluster bootstrap samples, resampling participants independently within each laboratory while retaining all their observations. The primary families were fixed before analysis as NREM spectral power and REM filtered Catch22. Frozen sensitivities included participant-equal weighting, site macro weighting, REM with Turku excluded, and the full pre-existing feature set.

3 Results

3.1 Approximate benchmark recovery

The adapted workflow recovered the scale and direction of the published benchmark. NREM spectral power reached record-wise AUROC = 0.569 (DREAM: 0.586); REM filtered Catch22 reached AUROC = 0.636 (DREAM: 0.700). The discrepancy is consistent with the missing feature module and unavailable Siclari preprocessing, so these values are treated as an approximate code-path reproduction.

Earlier fold-mean participant-held-out summaries were AUROC = 0.550 and 0.663. Reconstructing record-level out-of-fold probabilities allowed estimands to be evaluated on a common prediction set. Pooled participant-held-out AUROC was 0.534 [0.500, 0.569] for NREM spectral power and 0.629 [0.567, 0.701] for REM filtered Catch22. The change from the earlier summaries reflects fold-mean versus pooled-pair weighting, not retraining on different records.

3.2 Pooled and site-conditioned performance diverge

Restricting comparisons to records from the same laboratory reduced the primary pair-weighted AUROCs to 0.498 [0.462, 0.536] for NREM and 0.498 [0.440, 0.576] for REM. Equal-laboratory macro estimates were 0.510 [0.461, 0.559] and 0.498 [0.439, 0.595]. Thus, the primary pooled estimates did not describe a common within-laboratory ordering.

Results were heterogeneous rather than uniformly null. NREM spectral power was above chance in Zhang but near or below chance in the other collections; REM filtered Catch22 was below chance in the extremely imbalanced Turku collection and imprecisely above chance in

Table 1: Stage-specific analytic composition. Experience prevalence is computed after all exclusions. Participants contributing both stages are counted in each stage. *Fewer than five No-experience records; within-site AUROC is correspondingly fragile.

Stage	Collection	Participants	Records	Experience	No experience	Experience (%)
NREM	Young Adults	32	32	18	14	56.2%
NREM	Multiple Awakenings	19	281	189	92	67.3%
NREM	Siclari	39	231	124	107	53.7%
NREM	Zhang	27	186	139	47	74.7%
REM	Young Adults	30	30	20	10	66.7%
REM	Multiple Awakenings	19	154	125	29	81.2%
REM	Zhang	20	26	17	9	65.4%
REM	Turku*	18	125	122	3	97.6%

Zhang (Supplementary Table 2). The fold-conditioned pair-weighted sensitivity was 0.509 [0.471, 0.551] for NREM and 0.552 [0.501, 0.643] for REM. The corresponding differences from the 0.5 site-only baseline were 0.009 [-0.029, 0.051] and 0.052 [0.001, 0.143]. Excluding Turku changed the REM fold-conditioned estimate to 0.563 [0.510, 0.659], confirming sensitivity to collection composition rather than a stable common effect.

3.3 Site-only prediction and pair decomposition

Using no EEG features, training-fold site prevalence produced pooled AUROCs of 0.564 [0.511, 0.616] in NREM and 0.687 [0.611, 0.767] in REM. This does not imply that the EEG model is equivalent to the baseline; it shows that laboratory membership alone carries substantial pooled ranking information.

The pair decomposition identifies where the primary pooled AUROC originates (Figure 1). Within-site pairs comprised only 30.6% of NREM and 30.0% of REM pairs. Their concordances were 0.498 and 0.498, whereas cross-site concordances were 0.550 and 0.685. The weighted within- and cross-site contributions reconstruct the pooled AUROC to machine precision. Directional off-diagonal cells also reveal that a score’s ranking can reverse with the source pairing.

3.4 Performance does not transport to every laboratory

With an entire laboratory held out, the primary macro AUROCs were 0.493 for NREM spectral power and 0.470 for REM filtered Catch22; participant-equal summaries were 0.484 and 0.454. Target estimates varied widely (Figure 2), and no primary model had a favorable participant-cluster interval excluding 0.5 in every target. Turku contains only three No-experience records, making its REM interval and any aggregate that includes it especially unstable.

3.5 Laboratory identity and mechanistic controls

Laboratory identity remained strongly decodable under participant-held-out evaluation. Across PSD, broadband Catch22, and filtered Catch22, macro AUROC ranged from 0.808 to 0.953. This establishes large residual domain information but does not prove that every experience prediction is artifactual.

Neither advanced representation supplied a robust rescue. Riemannian covariance and label-free log-Euclidean alignment remained heterogeneous across laboratories (Supplementary Table 3). Raw temporal irreversibility was the strongest exploratory REM family under laboratory shift (AUROC = 0.604), but fell to AUROC = 0.490 after normalization against phase-randomized surrogates. Because those surrogates preserve each channel’s spectrum while disrupting nonlinear temporal ordering, this reduction does not support a domain-invariant arrow-of-time interpretation for the tested statistics.

4 Discussion

The principal result is a separation of estimands. The released workflow approximately reproduces pooled record-wise discriminability. Some pooled participant-held-out discrimination remains when people are separated. Yet the two primary models are approximately chance when positive-negative pairs are restricted to the same laboratory, and their decision rules do not transport consistently to an unseen laboratory. Participant-held-out validation therefore preserved an apparently successful multi-center benchmark while leaving laboratory-level structure shared across training and test data.

The site-only baseline and pair decomposition clarify, but do not fully causally explain, that divergence. About seven in ten pooled pairs were cross-laboratory comparisons. Such comparisons can reward stable differences in score location that covary with laboratory prevalence, acquisition, or reporting protocol without requiring within-laboratory experience ranking. Conversely, the fold-conditioned REM sensitivity was modestly above

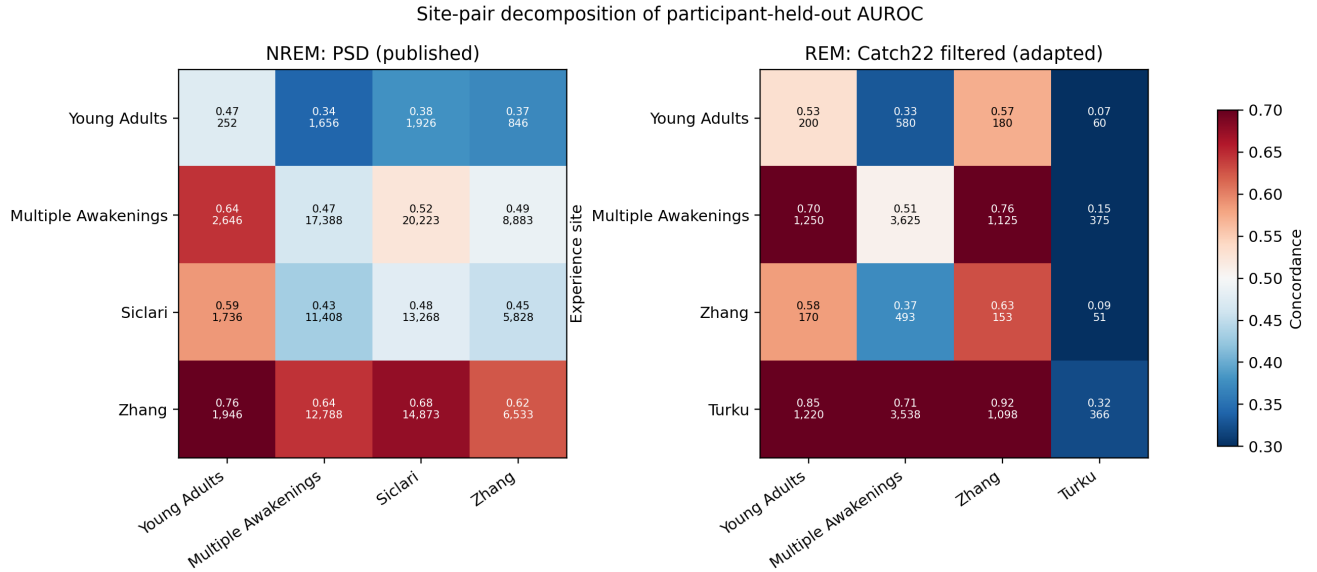


Figure 1: Exact site-pair decomposition of participant-held-out pooled AUROC for the two primary models. Rows identify the source of Experience records and columns the source of No-experience records. Each cell reports concordance (top) and the number of positive-negative pairs (bottom), averaged over the fixed repeated folds where appropriate. Diagonal cells are within-site; off-diagonal cells are cross-site.

0.5 and increased after excluding Turku. The appropriate interpretation is therefore heterogeneity: the pooled benchmark combines collection-specific effects, and a reproducible within-domain signal may exist in some settings, but a common transportable marker has not been established.

The mathematical extensions are informative chiefly as falsification tests. Riemannian geometry is principled for covariance-domain shift, but only three common channels with heterogeneous references were available. Log-Euclidean centering removes a domain mean, not arbitrary changes in covariance or label semantics. Temporal irreversibility has a plausible link to conscious dynamics, yet the phase-surrogate control shows why mechanistic null models are necessary before attributing performance to nonlinear temporal direction.

These findings do not show that EEG lacks correlates of dreaming. The labels themselves are imperfect proxies: “No experience” may combine absent experience, failed encoding, forgetting, and protocol-specific categorization. Failure across collections can therefore reflect genuine construct heterogeneity as well as instrumentation and pre-processing shift. A stronger future design would prospectively harmonize channels, reference, awakenings, question wording, and class balance, then reserve laboratories for external validation from the outset.

4.1 Limitations

This reanalysis was exploratory rather than preregistered; the targeted diagnostic protocol was frozen only before the new site-conditioned runs. The missing feature module and reconstructed Siclari channels preclude exact nu-

merical reproduction. Each stage has only four eligible target laboratories, and Turku is severely imbalanced. Analyses used three common channels and the final 30 s only. Participant-cluster bootstrap intervals do not capture uncertainty over a wider population of laboratories. Log-Euclidean alignment is transductive because it uses the unlabeled target batch. Finally, the novelty search was focused, not a registered systematic review; to our knowledge, we did not identify an earlier laboratory-held-out evaluation of this exact workflow.

5 Conclusion

Across the tested DREAM collections, the released Figure 5 workflow supports pooled discriminability but does not yet establish a transportable EEG marker of reported dream experience. Participant separation, site-conditioned ranking, site-prevalence baselines, dataset-identity controls, and unseen-laboratory validation should be reported as distinct tests in multicenter dream decoding.

Data and code availability

The public reproducibility repository at github.com/asim-v/dream-decoder-generalization-paper contains the frozen feature matrices, deterministic folds, complete out-of-fold probabilities, machine-readable results, tests, table and figure generators, environment lockfile, LaTeX source, and compiled PDF. One top-level command recreates the diagnostics and manuscript assets from the redistributed derived features. Raw EEG is not

Laboratory-held-out performance by target collection
Solid and dotted vertical lines mark participant-equal and record-weighted macro means

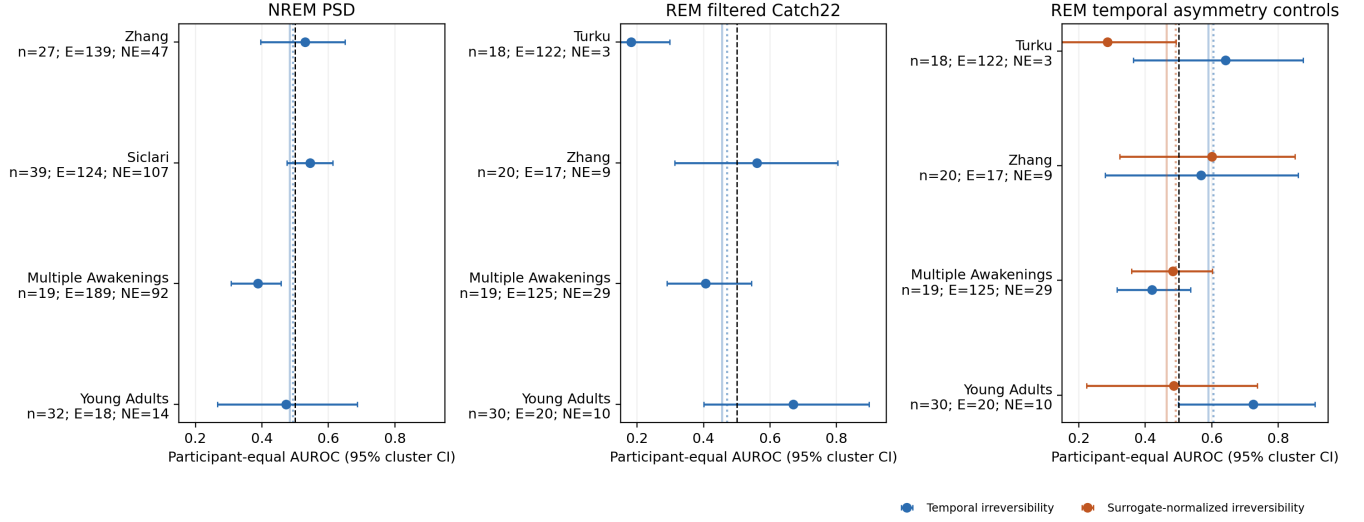


Figure 2: Laboratory-held-out participant-equal AUROC with 95% participant- cluster bootstrap intervals. The first two panels show the fixed primary models. The third shows raw temporal irreversibility (blue) and its phase- surrogate-normalized control (orange). Solid and dotted vertical lines are participant-equal and record-weighted macro means. E and NE denote Experience and No-experience records.

redistributed; the README provides persistent source identifiers, sizes, hashes, and acquisition instructions. Exact reproduction of the upstream Figure 5 feature matrix remains blocked by the unreleased `feat_ext_funcs` module and preformatted Siclari files expected by the deposit.

Ethics statement

This study reanalyzed de-identified public data and did not recruit participants or modify source records. Ethical approvals and consent procedures are described in the originating studies and DREAM records.

Competing interests

The author declares no competing interests.

A Supplementary results

Table 2: Primary participant-held-out within-site AUROC by collection. Values use record-weighted pairs; intervals resample participants within collection. E and NE are the two strict report classes.

Stage	Collection	Participants	E	NE	AUROC [95% CI]
NREM	Young Adults	32	18	14	0.474 [0.313, 0.620]
NREM	Multiple Awakenings	19	189	92	0.465 [0.409, 0.520]
NREM	Siclari	39	124	107	0.478 [0.421, 0.539]
NREM	Zhang	27	139	47	0.625 [0.533, 0.706]
REM	Young Adults	30	20	10	0.532 [0.356, 0.711]
REM	Multiple Awakenings	19	125	29	0.508 [0.444, 0.598]
REM	Zhang	20	17	9	0.629 [0.467, 0.783]
REM	Turku	18	122	3	0.322 [0.237, 0.462]

Table 3: Complete frozen feature-family sensitivity: participant-held-out, within-site pair-weighted AUROC. These families were not selected by the new diagnostic results.

Stage	Feature family	AUROC [95% CI]
NREM	PSD (published)	0.498 [0.462, 0.536]
NREM	Catch22 broadband (published)	0.479 [0.448, 0.510]
NREM	Catch22 filtered (adapted)	0.502 [0.466, 0.537]
NREM	Riemannian covariance	0.464 [0.433, 0.499]
NREM	Log-Euclidean domain alignment	0.463 [0.435, 0.497]
NREM	Temporal irreversibility	0.508 [0.473, 0.542]
NREM	Surrogate-normalized irreversibility	0.525 [0.495, 0.555]
REM	PSD (published)	0.458 [0.375, 0.532]
REM	Catch22 broadband (published)	0.475 [0.391, 0.559]
REM	Catch22 filtered (adapted)	0.498 [0.440, 0.576]
REM	Riemannian covariance	0.520 [0.449, 0.601]
REM	Log-Euclidean domain alignment	0.512 [0.442, 0.578]
REM	Temporal irreversibility	0.565 [0.482, 0.653]
REM	Surrogate-normalized irreversibility	0.598 [0.494, 0.693]

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